

Carlos Pineda-Antunez

Curriculum Vitae

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Carlos Pineda-Antunez

PhD student in Health Economics and Outcomes Research
Comparative Health Outcomes, Policy, and Economics Institute, University of Washington
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Research Profile

Health economist and decision scientist focused on health economics, outcomes research, decision modeling, and public health policy. My work evaluates prevention interventions, health system performance, and the value of health technologies, with applications in colorectal cancer, HIV, maternal health, mental health, and financial protection. I develop decision models, simulation tools, economic evaluations, and reproducible software workflows to support evidence-informed decision making.

Education

University of Washington, Seattle, WA, USA
PhD in Health Economics and Outcomes Research, expected 2027
National Institute of Public Health, Morelos, Mexico
MS in Health Economics, 2015
National Polytechnic Institute, Mexico City, Mexico
BS in Biomedical Engineering, 2011

Professional Experience

Stanford University, California, USA
Decision Modeler, Department of Health Policy, School of Medicine, 2025-current
University of Washington, Seattle, WA, USA
Teaching Assistant, Data Science in Health and Outcomes Research Certificate Program, Comparative Health Outcomes, Policy, and Economics Institute, School of Pharmacy, 2025-2026
University of Minnesota, Minneapolis, MN, USA
Health Economist, Division of Health Policy and Management, School of Public Health, 2024-2025
University of Washington, Seattle, WA, USA
Health Economist, Comparative Health Outcomes, Policy, and Economics Institute, School of Pharmacy, 2024-2025
Stanford University, California, USA
Decision Modeler, Department of Health Policy, School of Medicine, June-September 2024
Health Research Consortium (CISIDAT), Morelos, Mexico
Decision Scientist, Division of Quantitative Methods, 2022-2024
Center for Research and Teaching in Economics (CIDE), Mexico City, Mexico
Decision Modeler, Division of Public Administration, 2021-2022
National Institute of Public Health (INSP), Morelos, Mexico
Health Economist, Division of Health Economics and Health Systems Innovations, 2015-2021

Software Development and Published Tools

BayCANN for CISNET colorectal cancer models

Code to perform emulator-based Bayesian calibration for the three Colorectal Cancer CISNET models. Publication: [Medical Decision Making](#). Repository: [NCI-CISNET-Colorectal/baycann_cisnet_crc](#).

Cancer modeling with discrete-event simulation

Tutorial and open-source workflow for cancer modeling with discrete-event simulation. Publication: [PharmacoEconomics](#). Repository: [sjpi22/tutorial_cancer_modeling_des](#).

ggpop

R package built on top of ggplot2 to simplify the creation of icon-based population charts. CRAN page: [ggpop](#). Project website: [jurjoraa.github.io/ggpop](#).

Publications

- Servan-Mori E, Cerecero-Garcia D, **Pineda-Antunez C**, Jan S, Hone T, Flamand L, Mohar-Betancourt A, Mendoza MA, Mendez J, Garcia-Diaz R, et al. (2026). Health financing inequality in fragmented health systems: evidence from Mexico, 2000-2023. *International Journal for Equity in Health*.
- Pi S, Rutter CM, **Pineda-Antunez C**, Chen JH, Goldhaber-Fiebert JD, Alarid-Escudero F. (2026). Discrete-Event Simulation Modeling Framework for Cancer Interventions and Population Health in R (DESCIPHR): An Open-Source Pipeline. *PharmacoEconomics*. 1-19.
- Cerecero-Garcia D, Gomez-Dantes O, Hone T, **Pineda-Antunez C**, Mohar-Betancourt A, Flamand L, Servan-Mori E. (2026). Catastrophic and impoverishing health expenditures in fragmented public health systems: lessons from Mexico, 2000-2022. *Health Economics Review*.
- **Pineda-Antunez C**, Seguin C, van Duuren LA, et al. (2024). Emulator-Based Bayesian Calibration of the CISNET Colorectal Cancer Models. *Medical Decision Making*. 44(5):543-553.
- Servan-Mori E, Quezada-Sanchez AD, Fuentes-Rivera E, **Pineda-Antunez C**, Hernandez-Chavez M del C, Garcia-Martinez A, Garcia-Feregrino R, Madrigal A, Guerrero B, Medrano G, Schnaas L (2023). Proximal determinants of suboptimal early child development during the first three years of life in socially deprived Mexican contexts. *PLoS ONE*. 18(11).
- **Pineda-Antunez C**, Seguin C, van Duuren L, Knudsen AB, Davidi B, Nascimento De Lima P, Rutter C, Kuntz KM, Lansdorp-Vogelaar I, Collier N, Ozik J, Alarid-Escudero F (2023). Emulator-based Bayesian calibration of the CISNET colorectal cancer models. *Preprint*.
- Servan-Mori E, Fuentes-Rivera E, Quezada AD, **Pineda-Antunez C**, Hernandez-Chavez M del C, Garcia-Martinez A, Madrigal A, Garcia-Feregrino R, Santiago-Angelino T, Hernandez-Serrato M, Schnaas L (2022). Early neurological development and nutritional status in Mexican socially deprived contexts. *PLoS ONE*. 17(6).
- Quezada-Sanchez AD, Fuentes-Rivera E, Garcia-Martinez A, Hernandez-Chavez M del C, **Pineda-Antunez C**, Servan-Mori E (2022). Assessing the effect of an educational intervention on early childhood development among Mexican preschool children in the state of Oaxaca: a study protocol of a cluster randomized stepped-wedge trial. *Trials*. 23(1):126.
- Servan-Mori E, **Pineda-Antunez C**, Bravo-Ruiz ML, Molina M, Ramirez-Baca MI, Garcia-Martinez A, Quezada-Sanchez AD, Orozco-Nunez E (2022). A behavioral economics analysis of the participation in early childhood development social programs promoted by civil societies in Mexico. *PLoS ONE*. 17(3):e0265389.
- Bautista-Arredondo S, **Pineda-Antunez C**, Cerecero-Garcia D, Cameron DB, Alexander L, Chiwevu C, Levin C (2021). Moving away from the “unit cost”: Predicting country-specific average cost curves of VMMC services accounting for variations in service delivery platforms in sub-Saharan Africa. *PLoS ONE*. 16(4).
- Diaz-Castro L, Cabello-Rangel H, **Pineda-Antunez C**, Perez de Leon A (2021). Incidence of catastrophic healthcare expenditure and its main determinants in Mexican households caring for a person with a mental disorder. *Global Mental Health*. 8.
- Colchero MA, Gomez R, Pineda-Antunez CJ, Bautista-Arredondo S (2021). Health care utilization during the Covid-19 pandemic in Mexico: the cascade of care. *Salud Publica de Mexico*. 63(6):743-750.
- **Pineda-Antunez C**, Contreras-Loya D, Rodriguez-Atristain A, Opuni M, Bautista-Arredondo S (2021). Characterizing health care provider knowledge: Evidence from HIV services in Kenya, Rwanda, South Africa, and Zambia. *PLoS ONE*. 16(12).
- Siapka M, Vassall A, Cunnam L, **Pineda-Antunez C**, Cerecero D, Sweeney S, Gomez GB (2020). Cost of tuberculosis treatment in low- and middle-income countries: Systematic review and meta-regression. *International Journal of Tuberculosis and Lung Disease*. 24(8):802-810.
- Cabello-Rangel H, Diaz-Castro L, **Pineda-Antunez C** (2020). Cost-Effectiveness Analysis of Interventions to

- Achieve Universal Health Coverage for Schizophrenia in Mexico. *Salud Mental*. 43(2):249-255.
- Cameron DB, Mustafa Diab M, Carroll LN, Bollinger LA, DeCormier Plosky P, **Pineda-Antunez C**, Levin C, Kahn JG (2019). The state of costing research for HIV interventions in sub-Saharan Africa. *African Journal of AIDS Research*. 18(4):277-288.
 - **Pineda-Antunez C**, Martinez-Silva G, Cerecero-Garcia D, Alexander L, Cameron DB, Chiwevu C, Bautista-Arredondo S (2019). Meta-analysis of average costs of HIV testing and counselling and voluntary medical male circumcision across thirteen countries. *African Journal of AIDS Research*.
 - Servan-Mori E, Cerecero-Garcia D, Heredia-Pi IB, **Pineda-Antunez C**, Sosa-Rubi SG, Nigenda G (2019). Improving the effective maternal-child health care coverage through synergies between supply and demand-side interventions: Evidence from Mexico. *Journal of Global Health*. 9(2):1-12.
 - Servan-Mori E, Mendoza MA, Chivardi C, **Pineda-Antunez C**, Rodriguez-Franco R, Nigenda G (2019). A spatio-temporal cluster analysis of technical efficiency in the production of outpatient maternal health services and its structural correlates in Mexico. *International Journal of Health Planning and Management*. 34(4):e1417-e1436.
 - Cerecero-Garcia D, **Pineda-Antunez C**, Alexander L, Cameron D, Martinez-Silva G, Obure CD, Bautista-Arredondo S (2019). A meta-analysis approach for estimating average unit costs for ART using pooled facility-level primary data from African countries. *African Journal of AIDS Research*.
 - Rodriguez-Franco R, Servan-Mori E, Gomez-Dantes O, Contreras-Loya D, **Pineda-Antunez C** (2018). Old principles, persisting challenges: Maternal health care market alignment in Mexico in the search for UHC. *PLoS ONE*. 13(7):1-14.
 - Diaz-Castro L, Cabello-Rangel H, Arredondo A, Madrigal de Leon E, **Pineda-Antunez C** (2017). Cost-effectiveness of therapeutic interventions in schizophrenia. *The European Journal of Psychiatry*. 31(1):11-16.

Talks

- 2024. Quantifying the value of a potential new diagnostic test in the context of imperfect adherence. Conference presentation, Society for Medical Decision Making Annual Meeting, Boston, MA, USA.
- 2024. Fragmentation of health systems and maternal mortality transition in Mexico, 2000-2021. Conference presentation, Health Systems Global, Mexico City, Mexico.
- 2023. Optimizing a natural history simulation model of colorectal cancer using GPUs. Conference presentation, Society for Medical Decision Making Annual Meeting, Philadelphia, PA, USA.
- 2022. Emulator-based Bayesian calibration of CISNET colorectal cancer models. Poster and oral presentation, Society for Medical Decision Making Annual Meeting, Seattle, WA, USA.
- 2021. The Economic Impact of COVID-19 Non-Pharmaceutical Interventions and the influence of Poverty and Inequality. Oral presentation, International Health Economics Association, Virtual congress.
- 2020. Policy, poverty, and politics: How public health, inequality and political leadership interacted in the COVID-19 pandemic. Conference presentation, IAS COVID-19 Conference, San Francisco, CA, USA.
- 2019. Exploring the intervention costs of PMTCT and HTC: explaining and predicting unit costs for implementation scenarios using pooled data from several studies. Conference presentation, International Health Economics Association, Basel, Switzerland.
- 2018. Filling in the gaps: Analysis of VMMC unit costs using aggregated facility-level primary data from several studies. Conference presentation, International AIDS Conference, Amsterdam, Netherlands.

Teaching

University of Washington, CHOICE Grad Seminar
Course Master, March-June 2026

University of Washington, Advanced Methods in Economic and Outcomes Evaluation in Health and Medicine
Guest Lecturer, February 2026

University of Washington, Data Science in Health Economics and Outcomes Research Certificate Program
Teaching Assistant, September 2025-June 2026

National Institute of Public Health, Health Economics and Economic Evaluation courses
Teaching Assistant and committee service, 2016-2021

Awards

2023, Rubenstein Endowed Memorial Scholarship, University of Washington

2023, Graduate School Top Scholar Fellowship, University of Washington
2023, CONACyT Graduate Fellowship for pursuing a PhD program abroad
2022, SMDM Lee B. Lusted Student Prize finalist in Quantitative Methods and Theoretical Developments
2015, Thesis honorable mention, National Institute of Public Health
2013, CONACyT Graduate Fellowship for master's studies
2009, Academic excellence recognition in undergraduate studies in biomedical engineering

Skills

Economic evaluation; cost-effectiveness analysis; budget impact analysis; value of information analysis; probabilistic sensitivity analysis; health decision modeling; Markov models; microsimulation; discrete-event simulation; dynamic transmission models; causal inference; impact evaluation; machine learning for impact evaluation; cost analysis; systematic reviews; meta-analysis; R; reproducible research workflows.